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Effect of eastern oysters (*Crassostrea virginica*) on sediment carbon and nitrogen dynamics in an urban estuary

Timothy J. Hoellein[1,3] and Chester B. Zarnoch[2]

[1]*Biology Department, Loyola University Chicago, 1032 West Sheridan Road, Chicago, Illinois 60660 USA*
[2]*Department of Natural Sciences, Baruch College, City University of New York, Box A-0506, 17 Lexington Avenue, New York, New York 10010 USA*

Abstract. Oyster reefs have declined globally. Interest in their restoration has motivated research into oyster-mediated ecosystem services including effects on biodiversity, filtration, and nitrogen (N) cycling. Recent evidence suggests oysters may promote denitrification, or anaerobic respiration of nitrate (NO_3^-) into di-nitrogen gas, via benthic deposition of carbon (C) and N-rich biodeposits. However, the mechanisms whereby biodeposits promote N transformations prerequisite to denitrification (e.g., mineralization and nitrification) are unclear. Previous research has also not measured oysters' influence on N cycling in urbanized areas. In May 2010 we deployed eastern oysters (*Crassostrea virginica*) in mesh cages above sand-filled boxes at four sites across a nutrient gradient in Jamaica Bay, New York City (New York, USA). Oysters were arranged at four densities: 0, 40, 85, and 150 oysters/m^2. For 17 months we measured water-column nutrients and chlorophyll *a*, every two weeks to monthly. Every two months we measured sediment ash-free dry mass (AFDM), exchangeable ammonium (NH_4^+), ammonification, nitrification, denitrification potential (DNP), and NO_3^- and C limitation of DNP. Oysters increased sediment AFDM at three of four sites, with the greatest increase at high density. Oysters did not affect any N pools or transformations. However, variation among sites and dates illustrated environmental drivers of C and N biogeochemistry in this urban estuary. Overall, nitrification was positively related to net ammonification, water column NH_4^+, and sediment NH_4^+, but was not correlated with DNP. Denitrification was consistently and strongly NO_3^- limited, while C was not limiting or secondarily limiting. Therefore, the oyster-mediated increase in AFDM did not affect DNP because C was not its primary driver. Also, because DNP was unrelated to nitrification, it is unlikely that biodeposit N was converted to NO_3^- for use as a denitrification substrate. Predicting times or sites where denitrification is driven by the C and N species originating from oyster biodeposits remains a challenge under eutrophic conditions. Towards this goal, we synthesized our conclusions with literature predictions in a conceptual model for pathways whereby oysters might influence C and N dynamics differently in oligotrophic relative to eutrophic ecosystems.

Key words: *biodeposits; bivalves;* Crassotrea virginica; *denitrification; ecosystem services; eutrophication; Jamaica Bay, New York City, New York, USA*; *nitrification; oyster-reef habitat; restoration; urbanized coastal environment*.

Introduction

Oysters are ecosystem engineers, and their reefs provide multiple ecosystem services including enhanced biodiversity, sustainable fisheries, and water filtration (Newell 2004, Coen et al. 2007, Grabowski and Peterson 2007). Approximately 85% of oyster reefs have been lost worldwide (Beck et al. 2011), and the United States has lost ~88% of oyster biomass since the late 1800s (Zu Ermgassen et al. 2012). In general, loss of oysters and reefs has been attributed to overharvest, disease, reduced water quality, and modified hydrology (Beck et al. 2011, Zu Ermgassen et al. 2012).

The decline of oyster reefs and the ecological and economic implications associated with their loss has led to efforts to restore oyster reef habitat. Most of these efforts have targeted fishery restoration as the main goal (Beck et al. 2011), but have had limited success (Mann and Powell 2007, Beck et al. 2011). Although oyster-mediated ecosystem services and associated socioeconomic benefits have been clearly defined (Coen et al. 2007, Grabowski and Peterson 2007), there have been few studies that quantify these services in eutrophic conditions typical of urbanized coastal environments. The need for such empirical measurements is apparent as humans become increasingly dependent on coastal ecosystems (Weinstein et al. 2007) and managers consider oyster reef restoration as a tool for addressing cultural eutrophication (Cerco and Noel 2007, Carmichael et al. 2012).

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[3] E-mail: thoellein@luc.edu

Oyster restoration may help maintain ecosystem health in eutrophic waters by enhancing water filtration and the removal of excess nitrogen (N) via denitrification (Cornwell et al. 1999, Newell et al. 2005, Coen et al. 2007). Like other bivalve suspension feeders (Bruesewitz et al. 2006), oysters deliver N and carbon (C) from the water column to the benthos as biodeposits (i.e., feces and psuedofeces; Newell 2004). Increased water clarity from bivalve filtration can stimulate growth of submerged aquatic vegetation and microphytobenthic flora, stabilizing the sediments and increasing habitat for benthic fauna (Newell et al. 2005, Strayer 2009). Nutrients and organic matter in biodeposits can be mineralized by sediment microbes (Newell 2004), then assimilated by plankton and macroalgae, taken up by aquatic vegetation and microphytobenthic flora, or removed entirely from the aquatic environment as dinitrogen gas (N_2) through denitrification (Cornwell et al. 1999).

Recent evidence suggests oysters stimulate denitrification rates (Piehler and Smyth 2011, Kellogg et al. 2013), but the environmental conditions under which the N in oyster biodeposits is converted into N_2 gas are unclear. The relative abundance of C, dissolved oxygen (DO), and N, both in the water column and the sediment, affect multiple, interconnected N transformations including mineralization, nitrification, and denitrification (Cornwell et al. 1999, Herbert 1999). Therefore, the C and N in oyster biodeposits could have several biogeochemical fates, dependent upon environmental conditions. For example, biodeposits may serve as a source of C for denitrification of ambient NO_3^- (i.e., direct denitrification), and/or the N in biodeposits could be mineralized to stimulate coupled nitrification–denitrification. In locations with sediments rich in organic C and high water-column nutrients, anoxia can inhibit nitrification. Where nitrification and denitrification are strongly linked, denitrification can be reduced by low nitrification (Newell et al. 2002, Seitzinger et al. 2006). However, in anoxic sediments with sulfide, the final enzymatic steps of denitrification can be directly inhibited, or other NO_3^--reducing pathways favored (Burgin and Hamilton 2007). At sites where DO penetrates below the sediment–water interface, coupled nitrification–denitrification is more likely (Strauss and Lamberti 2000, Risgaard-Petersen 2003), and increased sediment diffusion of water column NO_3^- can stimulate direct denitrification. Documenting environmental drivers of C and N dynamics in sediments with oyster biodeposits is needed to predict whether oyster restoration can affect denitrification as an ecosystem service in eutrophic habitats.

Following global trends, a historically productive fishery of eastern oysters (*Crassostrea virginica*) in the Hudson–Raritan Estuary (HRE) in New York and New Jersey, USA, was lost by the 1920s (Franz 1982). Overfishing, habitat loss, and pollution contributed to the population decline, but sewage was the most significant factor, as eutrophic conditions stimulated hypoxia (Franz 1982). Although still present in low numbers, oysters are considered "ecologically extinct" in the HRE since they are no longer a keystone species and do not provide ecosystem services (NOAA Fisheries Eastern Oyster Biological Review Team 2007). In the second half of the 1900s increased wastewater treatment and reduced industrial activity improved HRE water quality (Brosnan et al. 2006), and coincided with interest in reviving oyster populations. Recent studies show strong oyster survivorship in waters around New York City (NYC), suggesting that eutrophication and hypoxia are no longer the primary factors limiting oyster populations. Instead, the current lack of oysters is attributed to reduced habitat availability, low natural recruitment, and disease (Levinton et al. 2011, Zarnoch and Schreibman 2012).

Although water quality in the HRE has improved in recent decades, it still receives more N (290 g $N \cdot m^{-2} \cdot yr^{-1}$) than any other estuary in the world (Howarth et al. 2006). Jamaica Bay, an embayment within the HRE, receives 15 800 kg nitrogen (N) per day, 89% of which originates from four large wastewater treatment plants (Benotti et al. 2007). Similar nutrient-loading rates occur throughout the HRE, particularly in the East River and western Long Island Sound (O'Shea and Brosnan 2000). N loading enhances growth of phytoplankton and *Ulva* spp. (sea lettuce), and decomposition of macroalgae and plankton can reduce DO, especially in summer (O'Shea and Brosnan 2000, Howarth et al. 2006). Oyster restoration has recently begun throughout the HRE, however, the influence of oysters on sediment N biogeochemistry in urbanized coastal ecosystems has not been quantified.

Our objectives were to measure the influence of oyster density on sediment C and N dynamics in eutrophic Jamaica Bay, NYC, at four locations that span a nutrient-loading gradient. Our hypotheses were that sites with high nutrients and high oyster density will show (1) increased sediment organic-matter content (i.e., from settled plankton or oyster biodeposits, respectively), and (2) increased evidence for denitrification fueled by sediment organic matter and water column NO_3^- (i.e., direct denitrification), rather than coupled nitrification–denitrification.

Methods

Study sites and experimental design

Jamaica Bay is a highly urbanized 5260-ha embayment on the southwestern coast of Long Island (New York, USA) with a relatively narrow inlet to the Atlantic Ocean (Fig. 1). The average depth is 5 m and there are multiple salt marsh islands throughout. Residence time ranges from 1 to 40 days from the inlet to the back of the bay, respectively. High N loading causes blooms of *Ulva* spp. and phytoplankton (i.e., chlorophyll $a > 100$ μg/L), and periodic anoxia (Benotti et al. 2007). Due to long-standing harvest restrictions,

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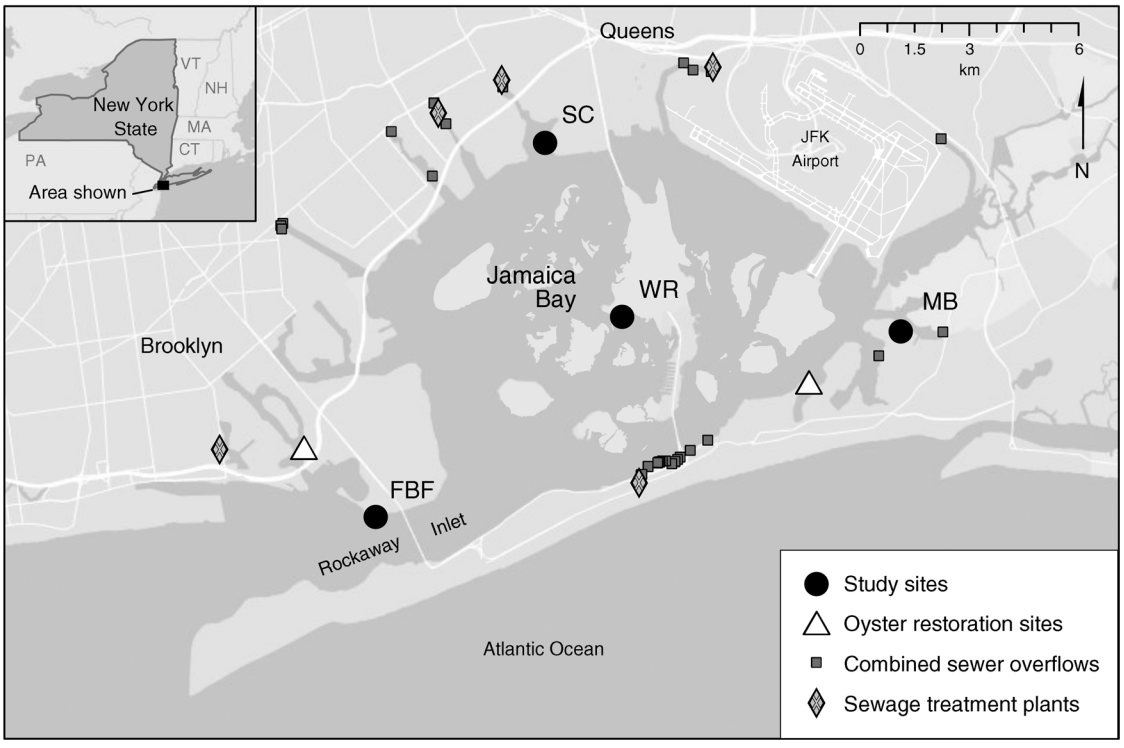


FIG. 1. The four study sites in Jamaica Bay, which is located in the boroughs of Brooklyn and Queens within New York City, New York, USA. The area is densely populated and has multiple combined sewer overflows and treatment plants. The study sites (Floyd Bennett Field [FBF], Wildlife Refuge [WR], Mott's Basin [MB], and Spring Creek [SC]) spanned a gradient of nutrient loading.

the bay supports large populations of northern quahogs (*Mercenaria mercenaria*) and soft-shell clams (*Mya arenaria*). Extremely dense ribbed mussel (*Geukensia demissa*) populations are also present in the bay (Franz 2001). Oysters were historically abundant, but no wild populations currently exist in the bay. Government-led oyster restoration is underway at two sites (Fig. 1).

We selected four sites in Jamaica Bay that represented a gradient of water quality conditions for our experiment. From lowest to highest N loading: Floyd Bennett Field (FBF) is at the bay's inlet, Motts Basin (MB) is in its southeastern corner, Jamaica Bay Wildlife Refuge (WR) is in the center, and Spring Creek (SC) is in the north center (Fig. 1). The SC site is adjacent to a water pollution control plant, which stores and releases treated wastewater. Numerous combined sewer overflows are located throughout the bay with a few situated near MB and SC (Fig. 1). At each site the location for oyster deployment was based on the amount of human activity (i.e., fewest anglers and foot traffic) and stable benthic surfaces.

We deployed oysters at four densities per site. Five-year-old oysters (shell length = 86.1 ± 9.1 mm [mean ± SD]; tissue dry mass = 2.75 ± 1.01 g), obtained from Frank M. Flower and Sons (Bayville, New York, USA) were placed on their left valve, evenly distributed within 27.5 × 30 × 10 cm polyethylene cages with 16-mm aperture openings. Densities were 40 oysters/m^2 (low), 85 oysters/m^2 (medium), and 150 oysters/m^2 (high), or none (control). Densities were based on measurements from natural reefs in Great Bay, New Hampshire, USA, which had a mean of 57 oysters/m^2 and a maximum of 157 oysters/m^2 (Capone et al. 2008), and restored reefs in Chesapeake Bay, which had mean densities of 173 oysters/m^2 (Rodney and Paynter 2006). Cages were attached directly on top of 660-cm^2 plastic boxes that were filled with 11 kg of Pavestone play sand (Pavestone Company, Atlanta, Georgia, USA). The distance between the sediment surface and the bottom of the cage was ~7 cm.

We deployed five replicate boxes of each treatment (N = 20) along with separate mesh bags of oysters for replacement of dead individuals. The boxes were individually attached to a trotline anchored with cinder blocks parallel to shore. Boxes were deployed subtidally at a depth of 1–1.5 m at mean low water and were accessible by wading only during spring tides. This was done to reduce vandalism, maximize submerged time of oysters, and allow access without a boat. Previous studies in Jamaica Bay showed this minimized vandalism and poaching (Zarnoch and Schreibman 2008, 2012). Cages were deployed on 26–27 May 2010, and were sampled each two to three months for sediment N dynamics through 31 October 2011.

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Our box + cage approach was designed to mimic oyster biodeposition under natural and aquaculture conditions. Under natural conditions oyster biodeposits settle on sediments adjacent to the reef. In aquaculture, biodeposits settle on underlying sediments. There were no oyster reefs or aquaculture in Jamaica Bay when this study began. We note the presence of the oysters in the cages relative to empty cages above the control boxes could affect water flow across sediments and particulate deposition. In addition, the porosity of the play sand could affect diffusion of solutes, and differences between it and natural substrates may reduce transferability of the findings. We chose play sand as the experimental substrate as it enabled uniformity among sites and for the experiment's duration, ensured no infaunal organisms were added or transferred among sites, and removed sediment inconsistency as a confounding factor. We address water flow and sediment effects further in the discussion.

Sampling procedure and schedule

On the collection dates, each box + cage was individually retrieved and transported by hand to the intertidal zone. Oyster cages were removed from the box to access sediment. We measured oyster mortality, replaced dead animals to maintain desired densities, and sampled the sediment. Cages were reattached to sediment boxes and redeployed within 2 h. Fouling organisms (i.e., macroalgae in early summer) occurred only at FBF and were removed with a hand brush prior to redeployment. Our sampling window was limited to 2–3 h low tides on only a few days each month, so we sampled one site our first day of field work, and another site the second day. This was followed by several days of laboratory work (see below). Therefore, each month we sampled sediment from two sites, and sampled the other two sites the following month.

We sampled sediment from four replicates of each of the four treatments. We obtained a composite sample of sediment from each box using a corer to remove the top 2–3 cm of sediment from three locations within the box, providing an aggregate sample of 150 mL of sediment, placed in 160-mL specimen containers. At FBF and SC (the least eutrophic and most eutrophic sites, respectively), we collected an additional 150 mL of sediment for nutrient-limitation assays of denitrification potential (DNP). We replaced the volume of sampled sediment with new sand distributed evenly across the sediment surface. This ensured the total volume of sediment in the boxes was constant throughout the entire experiment. This approach resulted in some disturbance or dilution of sediment; however, these effects were equal across all experimental units and likely minimal given the two-month window between sampling. We collected 3–7 L of unfiltered site water for nitrification and DNP measurements. Sediment samples and site water were placed in coolers on ice for transport to the laboratory. After the final sampling, a sediment subsample was preserved in formalin for later identification of meiofauna and macroinvertebrates.

We collected site water for measurements of inorganic N and phosphorus (P) and chlorophyll *a*, each two weeks to monthly. We filtered three 20-mL water column samples in the field (0.7-μm nominal pore size; Whatman GF/F [Whatman, Piscataway, New Jersey, USA), and froze samples until measurement of dissolved inorganic N and P. Simultaneously, we collected 2–3 L of water for chlorophyll *a* (chl *a*). Water was filtered on site or within 12 h and chl *a* filters were stored in dark containers at −20°C until processing (Parsons et al. 1984). Our sampling regime and distance among sites precluded water sampling on the same day or tidal cycle, which may have affected nutrient concentrations. However, salinity remained uniform at ~28 PSUs, as fresh water inflows are confined to wastewater inputs. In addition, all four sites were shallow and well-mixed with no evidence of water column dissolved oxygen (DO) stratification. Sampling was completed during daylight hours when DO was generally saturated due to high primary productivity.

Organic matter, exchangeable NH_4^+, net ammonification, nitrification, and C:N

Each sediment sample was thoroughly homogenized in its 160 mL specimen container prior to all measurements. Within 24–48 h of collection we initiated measurements of sediment ash-free dry mass (AFDM) and exchangeable ammonium (NH_4^+). We placed 5 mL of sediment onto a pre-ashed and weighed tin. Sediment was dried at 60°C for 3–7 days and ashed at 550°C for 3 h (Benfield 2006). A 2-mL subsample of the homogenized sediment was frozen at −20°C for analysis of organic C and N. Samples were dried at 60°C and acidified by two treatments of 25% HCl (Nieuwenhuize et al. 1994), redried at 60°C, and then measured on a Perkin Elmer 2400 Series II CHN Analyzer (PerkinElmer, Waltham, Massachusetts, USA). The C:N ratio was calculated as the mass ratio of the organic C to organic N in the sediment. To measure exchangeable NH_4^+, we placed 10 mL of sediment in a 50-mL centrifuge tube. Sediment was weighed, and an equal volume (mL) of 2 mol/L KCl was added (Sparks 2003). Centrifuge tubes were placed on shaker table at 150 rpm for 1 h. We centrifuged the samples at 6000 *g* ($1\ g = 9.80665\ m/s^2$) for 10 min, filtered the water (0.7-μm nominal pore size; Whatman GF/F), and filtrate was frozen until measurement of NH_4^+.

Within 24–48 h we initiated sediment nitrification assays using the nitrapyrin-inhibition method (Kemp et al. 1990, Strauss and Lamberti 2000). We added 25 mL of sediment and 50 mL of site water to each of two flasks. We added 20 μL of nitrapyrin + dimethyl sulfoxide (DMSO) to one, and 20 μL DMSO alone to the second. Sediments were incubated in the dark on a shaker table at 150 rpm to maintain aerobic conditions and constant light conditions. After 40–48 h, 25 mL of 2

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mol/L KCl was added to each flask, the sediments shaken for an additional 2 h, and then flasks were removed from the shaker table. We vigorously stirred each flask and removed 30 mL of the sediment slurry into a 50-mL centrifuge tube. Each sample was centrifuged, filtered, and frozen as described for exchangeable NH_4^+. Nitrification rate was calculated as the difference in NH_4^+ in the nitrapyrin + DMSO flask relative to DMSO-alone flask. We calculated net ammonification as the difference between initial exchangeable NH_4^+ concentration in the sediment and the exchangeable NH_4^+ concentration in the nitrapyrin treatment at the end of the incubation (Starry et al. 2005). This value is the net flux of NH_4^+ when nitrification is blocked, where a positive number indicates net NH_4^+ production (i.e., net ammonification), and a negative number net NH_4^+ uptake (i.e., net immobilization; Starry et al. 2005).

Denitrification potential and nutrient limitation

Rates of DNP were quantified using the chloramphenicol-amended acetylene (C_2H_2) inhibition technique, enriched with NO_3^- and glucose (Knowles 1990). We added 25 mL of sediment and 45 mL of unfiltered site water in 125-mL media bottles with butyl rubber septa. A solution of chloramphenicol (final concentration 0.3 mmol/L), NO_3^- (final concentration 6 mg N/L), and glucose (final concentration 30 mg C/L) was added to each sample (Bruesewitz et al. 2006). Chloramphenicol blocks de novo synthesis of new enzymes, preventing microbial adjustment to conditions in the assay bottles (Murray and Knowles 1999). The NO_3^- and glucose addition ensured denitrifying microbes have no nutrient limitation (Groffman et al. 2006). The bottles were purged for 5 min with ultra-high purity N_2 to establish anoxia, and returned to atmospheric pressure using a vent needle. We added 15 mL of C_2H_2 to the headspace to block conversion of nitrous oxide (N_2O) to N_2. After C_2H_2 addition, we collected 5 mL of gas from the bottles' headspace every 0.5–1 h for 4–5 h in pre-evacuated 3-mL containers. Following each collection, a 9:1 (per volume) mixture of N_2 to C_2H_2 was added to maintain constant pressure. Bottles were shaken by hand for 10 s before and after each sampling. All assays were conducted at room temperature. Nitrous oxide was measured using a gas chromatograph (GC 2014, Shimadzu Scientific Instruments, Columbia, Maryland, USA) equipped with electron capture detector (Groffman et al. 1999) and autosampler (AOC-5000; Shimadzu Scientific Instruments). We calculated DNP from the linear increase in N_2O (Murray and Knowles 1999). Ammonification, nitrification, and DNP were expressed as mass of N per surface area per hour.

Finally, we measured limitation of DNP by NO_3^- or C availability in sediment collected at the least eutrophic (FBF) and most eutrophic (SC) sites. DNP limitation assays were prepared identically to those described above, except we increased the number of replicates by a factor of 4 to include additional treatments. Our treatments included 5 mL of water that contained either (1) chloramphenicol alone (+0), (2) NO_3^- + chloramphenicol (+N), (3) glucose + chloramphenicol (+C), or (4) NO_3^- + glucose + chloramphenicol (+N+C; $N = 64$ assay bottles per site; Bruesewitz et al. 2006). We did not estimate C or N saturation of DNP, but preliminary data suggested no saturation in +N assays at either FBF or SC, despite relatively high water-column NO_3^- concentrations.

The C_2H_2 inhibition method has several limitations including inhibition of nitrification by C_2H_2 (which can deplete NO_3^- in assay bottles, especially if hydrogen sulfide is present), slow diffusion of C_2H_2 into sediments, and incomplete blockage of N_2O to N_2 (Cornwell et al. 1999, Groffman et al. 2006). In addition, the homogenized slurry used in this approach obscures in situ redox gradients. However, C_2H_2 inhibition remains widely used for measuring denitrification *potential*, allows for much higher replication at lower cost than isotope methods, and is useful for comparison among treatments in a spatially and temporally replicated experiment rather than for N budgets (Groffman et al. 2006). Rates of DNP in this study do not represent in situ denitrification rates, the values should not be scaled up to compare to other sites, and are not appropriate for use in N budgets here or elsewhere.

Water chemistry and chlorophyll a

We used standard spectrophotometric assays to measure nitrite (NO_2^-) + NO_3^- as NO_x^- (cadmium reduction; APHA 1998), soluble reactive phosphorous (SRP; antimonyl tartrate; Murphy and Riley 1962), and NH_4^+ (phenol hypochlorite; Solorzano 1969) using an Autoanalyzer II or AQ2+ (Seal Analytical, Mequon, Wisconsin, USA). Standard matrixes were adjusted to account for salinity and KCl. Chlorophyll *a* was determined spectrophotometrically following overnight extraction at 4°C in 90% acetone (Parsons et al. 1984).

Data analysis

We compared water chemistry and chl *a* using a two-way ANOVA by site and season. We condensed the data into four seasons because all samples were not collected on the same date across sites. We used solstice and equinox dates to separate seasons. We compared all response variables (i.e., sediment AFDM and C:N, exchangeable NH_4^+, net ammonification, nitrification, and DNP) among the four oyster treatments using a repeated-measures (RM) ANOVA at each site separately. When a significant effect of oyster density was found, we completed post hoc analyses using Fisher's least significant difference test. Finally, we were unable to include site as a factor in these analyses, as we collected data from two sites on alternate months. This schedule precluded simultaneous measurements across sites, but enabled collection of many more aspects of N and C biogeochemistry. Therefore, we compared results among

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TABLE 1. Results for repeated-measures ANOVA among dates and oyster density treatments for each study site for sediment carbon and nitrogen dynamics.

Site†	Oyster P‡	Date P	Interaction P
Sediment ash-free dry mass, AFDM			
FBF	**0.047**	**<0.001**	0.201
MB	0.272	**<0.001**	0.598
WR	**0.055**	**<0.001**	0.567
SC	**0.068**	**<0.001**	0.523
Sediment carbon-to-nitrogen ratio, C:N			
FBF	0.517	0.535	0.976
MB	0.298	**<0.001**	0.904
WR	0.088	**<0.001**	0.642
SC	0.519	0.380	0.895
Exchangeable ammonium, NH_4^+			
FBF	0.888	**<0.001**	0.773
MB	0.382	**<0.001**	0.087
WR	0.850	**0.006**	0.567
SC	0.752	**<0.001**	0.327
Net ammonification			
FBF	0.921	0.817	0.997
MB	0.959	**0.016**	0.812
WR	0.840	**<0.001**	0.074
SC	0.331	**0.056**	0.785
Nitrification			
FBF	0.206	0.402	0.324
MB	0.596	0.198	0.363
WR	0.844	0.113	0.495
SC	0.385	**0.002**	0.374
Denitrification potential, DNP			
FBF	0.216	**0.002**	**0.002**
MB	**0.058**	**<0.001**	0.256
WR	0.149	**<0.001**	0.481
SC	0.889	**<0.001**	**0.019**

Note: Boldface P indicates significance at $\alpha = 0.07$, which we chose given the high variation in biogeochemistry among treatments and dates.

† Sites in Jamaica Bay, New York City (New York, USA) are: FBF, Floyd Bennett Field; MB, Mott's Basin; WR, Wildlife Refuge; and SC, Spring Creek.

‡ "Oyster" refers to oyster density treatment (i.e., control, no oysters; low, 40 oysters/m^2; medium, 85 oysters/m^2; and high, 150 oysters/m^2).

sites by aligning the two-way RM ANOVA output for all four locations according to each response variable (Table 1).

We quantified nutrient limitation of DNP using two-factor ANOVA based on the presence or absence of N or C (Tank and Dodds 2003, Hoellein et al. 2011). N or C limitation was indicated when NO_3^- or C alone increased DNP without a significant interaction. We recorded co-limitation when NO_3^- and C each affected DNP, or when NO_3^-+C significantly increased DNP. We recorded primary N limitation with secondary C limitation if N alone increased DNP, and the combined NO_3^-+C treatment increased DNP even more (i.e., a significant N effect and a significant N × C interaction). We calculated the nutrient response ratio (NRR) as ln[DNP rate with +N+C/DNP rate with no added N or C] for each oyster treatment, and calculated the oyster response ratio (ORR) as ln[DNP oyster treatment/DNP control] (Hoellein et al. 2011). We used linear regression to compare the magnitude of DNP nutrient limitation to the magnitude of DNP oyster response.

To explain variation in nitrification and DNP among sites and dates, we used Akaike's information criterion (AIC) to select the best candidate models for multiple linear regression from hypothesized predictors. The lowest change in AIC (ΔAIC) indicated the best model, and models where ΔAIC < 2.0 were assumed equivalent (Anderson and Burnham 2002, Hall et al. 2009). For ANOVA and regression analyses, data were log-transformed to meet assumptions of equal variance and normal distribution if needed. All statistics were completed in SYSTAT 13 (Systat Software 2009).

RESULTS

Water chemistry and chlorophyll a

There was significant variation in nutrient and chl *a* concentrations among sites and seasons (Fig. 2). As expected, Spring Creek (SC) had the highest SRP (soluble reactive phosphorus), NH_4^+, and NO_x^- concentrations, Floyd Bennett Field (FBF) had the lowest, and Wildlife Refuge (WR) and Mott's Basin (MB) were intermediate (Fig. 2A–C). SRP and NH_4^+ concentrations were each lowest in winter and highest in summer, except NH_4^+ in Spring Creek showed minimal seasonal variation. In contrast, NO_x^- was lowest in summer and highest in autumn (Fig. 2C). Spring Creek maintained much higher NO_x^- in winter and spring than the other three sites. NO_2^- averaged ~20% of NO_x^-, with a greater relative contribution in spring and summer (data not shown). Spring Creek had higher values in chl *a* (ANOVA, $P = 0.063$; Fig. 2D). Across seasons, chl *a* was highest in winter and summer, and lowest in autumn. Oyster mortality was low (3.0% ± 1.1% [mean ± SD]) in winter, spring, and summer, but higher in autumn (33.0% ± 6.0%).

Oyster effect on sediment C and N dynamics

Oysters increased sediment ash-free dry mass (AFDM) in three out of the four study sites in Jamaica Bay, New York City (New York, USA) repeated-measures (RM) ANOVA, FBF $P = 0.047$, WR $P = 0.055$, SC $P = 0.068$; Fig. 3, Table 1). In all three sites the highest oyster density significantly increased AFDM over the control ($P < 0.05$ in all cases), while the low and medium densities were not significantly different from control or high density. There was no effect of oysters on sediment C:N or exchangeable NH_4^+ (Table 1), and both C:N and exchangeable NH_4^+ were variable through time. Oysters did not affect net ammonification or nitrification rates (Figs. 3 and 4, Table 1). While ammonification was variable across dates at all sites, there was no significant temporal variation in nitrification except for Spring Creek (RM ANOVA $P = 0.002$). Finally, there was no effect of oysters on DNP except at Mott's Basin, where DNP in the high oyster density was

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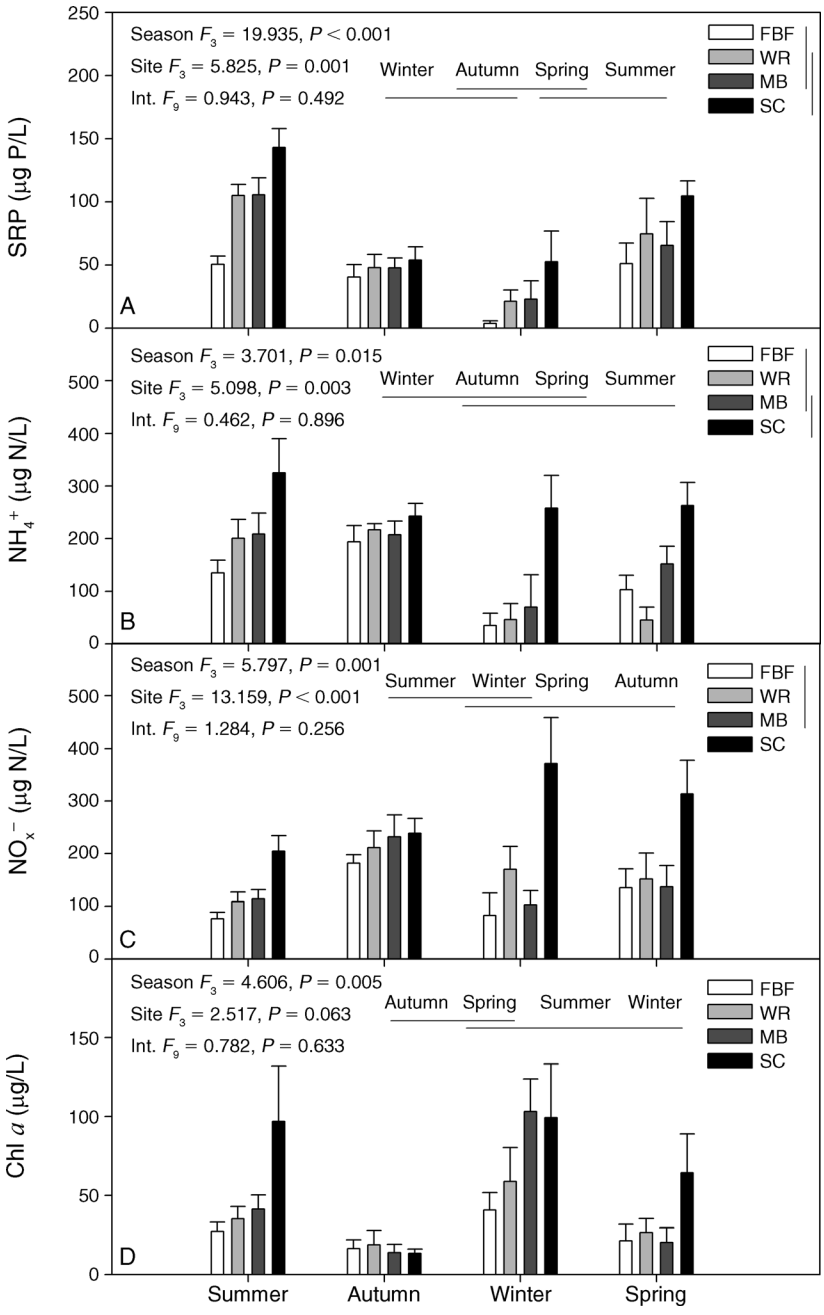


FIG. 2. Concentrations of (A) soluble reactive phosphorus (SRP), (B) ammonium (NH_4^+), (C) nitrate + nitrite (NO_x^-), and (D) chlorophyll *a* (chl *a*) across four seasons at the four sites (FBF, WR, MB, SC). The two-way ANOVA results by site and season ("Int.") are presented in the upper left corner of each panel. Significant differences (Tukey's multiple comparison) are shown for seasons by black lines above the histogram bars grouping seasons and for sites by black lines grouping sites in the site-code keys. The black lines group seasons and sites that are not significantly different at $\alpha = 0.05$. The data are means and SE. For site codes, see Fig. 1 legend.

higher than the control, but the low and medium density were not significantly different from the control or high density (RM ANOVA $P = 0.058$; Fig. 4, Table 1). Comparison of the patterns among the sites in Table 1 show unexpected uniformity across all four sites for all six of the response variables. The exception was Mott's Basin, where the influence of oysters was different from the other three sites for both sediment AFDM and DNP (Table 1).

Limitation assays of DNP showed significant NO_3^- limitation on every date at both the low-nutrient site (FBF) and high-nutrient site (SC; Table 2; $P < 0.001$ in

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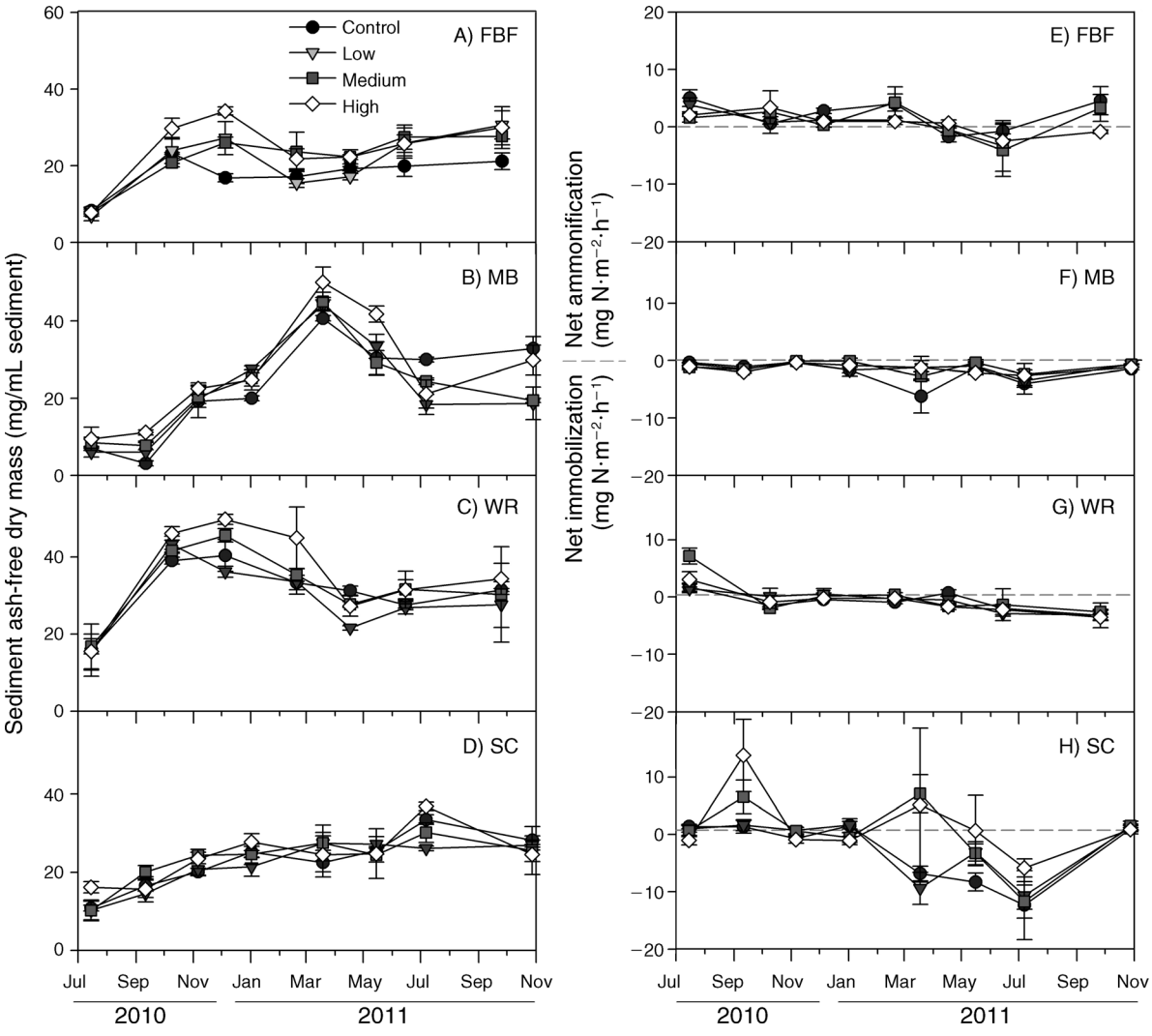


FIG. 3. Temporal variation in (A–D) sediment ash-free dray mass (AFDM) and (E–H) net ammonification/immobilization for four oyster treatments (control, low, medium, and high) at each of the four Jamaica Bay study sites. (On the *y*-axis, positive values are net ammonification, and negative values are net immobilization.) Results of repeated-measures ANOVA by oyster treatment, date, and treatment × date interaction are shown in Table 1. All sites were sampled in July 2010, but thereafter they were sampled in pairs, alternating pairs each month. The data are means + SE. For site codes, see Fig. 1 legend.

all cases). DNP was co-limited by NO_3^- and C in SC on three dates and in FBF on one date, and secondarily limited by C in FBF on three dates (Table 2). There was no difference in nutrient limitation patterns in the control sediment relative to the sediment exposed to oysters. However, the magnitude of change in DNP from the +N+C trials relative to the DNP trials with no nutrients added (i.e., nutrient response ratio, or ln[+N+C/+0]) was positively related to the change in DNP for the oyster-treatment sediments relative to the controls (i.e., oyster-treatment ratio, or ln[treatment/control]) (Fig. 5). That is, the influence of oysters on DNP was greatest when DNP was most strongly nutrient limited. Sediment from FBF in December 2010 was an outlier to this pattern (Fig. 5).

Environmental drivers of DNP and nitrification

AIC model selection with nitrification or DNP as the dependent variables showed each process was related to distinct physicochemical conditions (Table 3). Net ammonification rate, water column NH_4^+, and exchangeable NH_4^+ concentration explained 53.4% of variation in nitrification (nitrification model 5; Table 3), and the three factors together explained much more variation than models with ammonification or exchangeable NH_4^+ alone (nitrification models 1 and 2; Table 3). Of the two best models for nitrification, the most parsimonious also included water column SRP (nitrification model 8; Table 3). The equation for this model was nitrification = 468 + (0.332 × ammonification) + (0.181 × xNH_4^+) + (1.292 × NH_4^+) − (5.533 × SRP). In

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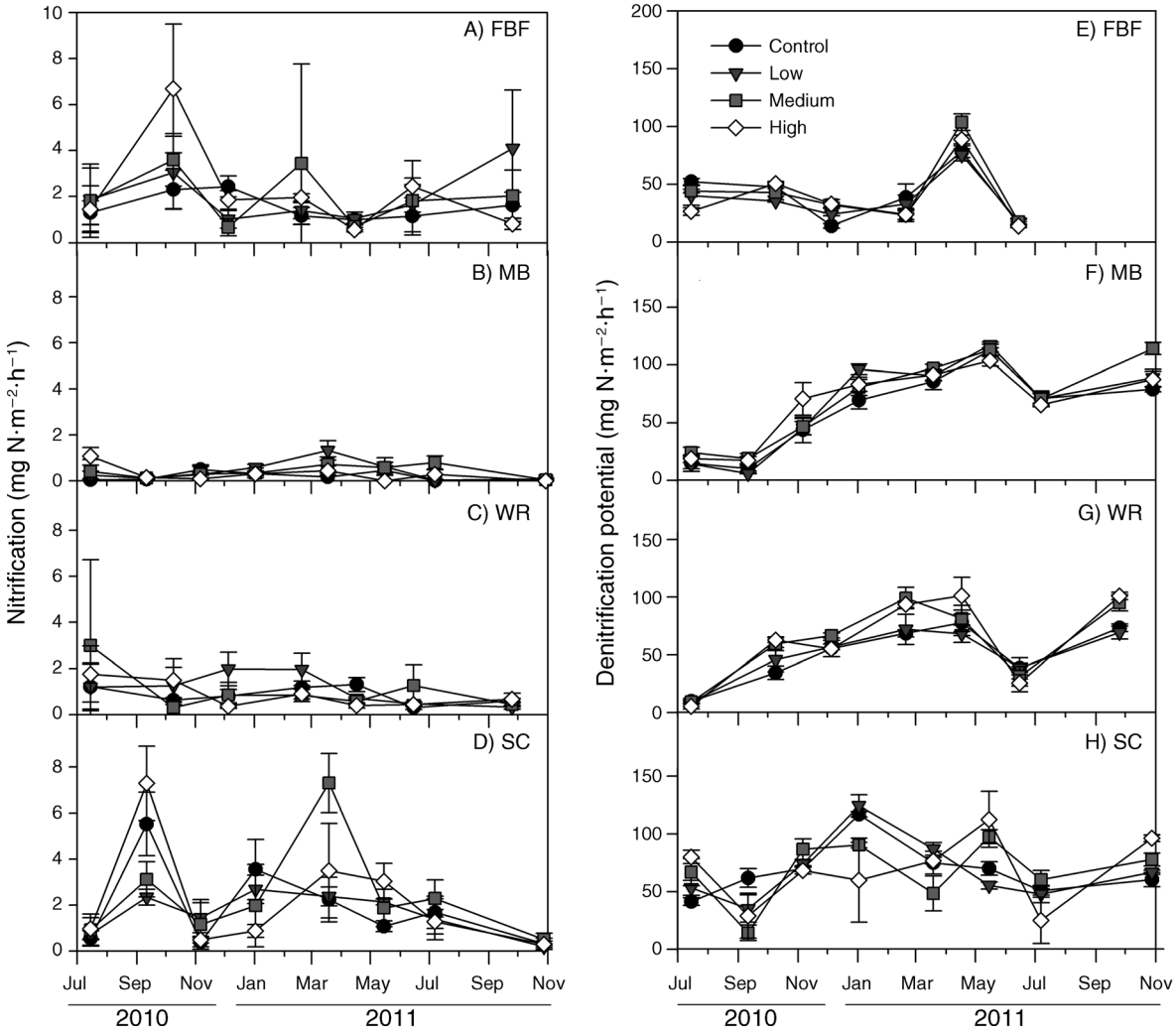


FIG. 4. Temporal variation in sediment (A–D) nitrification and (E–H) denitrification potential (DNP) for four oyster treatments (control, low, medium, and high) at each of the four Jamaica Bay study sites. Results of repeated-measures ANOVA by oyster treatment, date, and treatment × date interaction are shown in Table 1. All sites were sampled in July 2010, but thereafter they were sampled in pairs, alternating pairs each month. The data are means + SE. For site codes, see Fig. 1 legend.

contrast, 31.0% of variation in DNP was explained by sediment AFDM and water column NO_3^- together (DNP model 4; Table 3), but the best candidate model explained 38.8% of variation in DNP and also included chl *a*, water column NH_4^+, exchangeable NH_4^+, and sediment C:N (Table 3). The equation for this model was $\log(\text{DNP}) = 1.599 + (0.460 \times \text{AFDM}) + (0.002 \times NO_3^-) + (0.001 \times \text{Chl } a) - (0.001 \times \text{xNH}_4^+) - (0.001 \times NH_4^+) - (0.037 \times \text{C:N})$. Nitrification rate was not a significant predictor in any of the candidate models for DNP.

DISCUSSION

Oyster effect on sediment AFDM and DNP

The most consistent effect of oysters in this study was to increase sediment ash-free dry mass (AFDM), with a clear trend from zero to high oyster density at three of the four sites. Oyster biodeposits probably contributed to this difference as oysters were positioned in bags in such a way that sinking feces and pseudofeces could accumulate in the sediment. However, the presence of oysters in mesh cages may have enhanced seston settling rates into the boxes or reduced horizontal flow which could move accumulated organic matter out of the boxes. Mott's Basin was the only site to show no effect of oysters on AFDM. This site had higher AFDM in the high oyster treatment in year 1 (Fig. 3B), but not afterwards. This site also had some of the highest sediment AFDM values across sites. Of the four sites, Mott's Basin was best protected from wind fetch and was likely to have greater deposition and retention of phytoplankton and biodeposits in the sediments. High AFDM values across treatments may have obscured the

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TABLE 2. Nutrient limitation status of denitrification potential (DNP) at the low-nutrient site (FBF) and the high-nutrient site (SC) in Jamaica Bay, New York City.

Site and date	N	C	N × C	Limitation†
Spring Creek, SC				
15 Jul 2010	**<0.001**	0.426	0.291	N
11 Sep 2010	**<0.001**	0.839	0.539	N
6 Nov 2010	**<0.001**	**0.009**	0.897	N and C
2 Jan 2011	**<0.001**	0.725	0.633	N
19 Mar 2011	**<0.001**	0.801	0.708	N
15 May 2011	**<0.001**	**0.001**	0.776	N and C
7 Jul 2011	**<0.001**	0.876	0.910	N
28 Oct 2011	**<0.001**	**0.007**	0.704	N and C
Floyd Bennett Field, FBF				
11 Jul 2010	**<0.001**	0.536	**0.045**	N, 2° C
9 Oct 2010	**<0.001**	**0.018**	**0.023**	N and C
5 Dec 2010	**<0.001**	0.593	**<0.001**	N, 2° C
19 Feb 2011	**<0.001**	0.100	**<0.001**	N, 2° C
15 Apr 2011	**<0.001**	0.172	0.109	N
14 Jun 2011	**<0.001**	0.673	0.498	N

† Primary and secondary (2°) limitation were determined by additions of nitrate (N) and glucose (C) alone and together (ANOVA $P < 0.05$).

influence of biodeposit accumulation as the study progressed.

The increase in sediment AFDM with increasing oyster density had limited evidence for an effect of oysters on DNP. This is likely because DNP was explained by a combination of N and C availability. While water column NO_3^- concentration and sediment AFDM together were the strongest factors explaining variation in DNP (Table 3), DNP was limited more strongly and frequently by NO_3^- rather than by C (Table 2), and oysters increased DNP most when it was co-limited by NO_3^- and C (Fig. 5). An examination of the relationship between sediment AFDM and DNP alone showed the relationship seems to hold up to a threshold, above which DNP and organic matter were not related (Fig. 6). The data as a whole suggest that the increase in sediment AFDM only translates into increased rates of DNP when DNP is limited by C. Our results suggest that oysters are most likely to influence DNP at sites or time periods where it is co-limited by N or C.

While oysters increased sediment AFDM, organic N or NH_4^+ in their biodeposits could also contribute to N availability for DNP. To derive NO_3^- from oyster biodeposits, the N would require ammonification (from organic N to NH_4^+) and nitrification before being used as a substrate for denitrification (i.e., coupled nitrification–denitrification). Our data show no evidence oysters affected NH_4^+ pools or fluxes, as there was no influence of oysters on exchangeable NH_4^+, ammonification, or nitrification. The effect of oysters on sediment biogeochemistry was restricted to the C pool. Therefore, oysters influenced N cycling only where C and N transformations were linked, such as C limitation of DNP.

Unexpectedly, the two main factors of our experiment, oyster density and nutrient loading (i.e., site), did not affect the role of oysters on sediment N dynamics in this eutrophic ecosystem. The high density treatment of 150 oysters/m^2 had significantly more AFDM, but low and medium densities (40 and 85 oysters/m^2, respectively), showed no significant difference from the control. Historic values for oyster density in Jamaica Bay are unclear. We chose oyster density and size based on comparable estimates from other estuaries. For restoration purposes, our results suggest densities of large, mature oysters $\geq$150 oysters/m^2 may be required to affect sediment AFDM, while lower densities of smaller animals may be less likely to alter sediment C dynamics.

The range of physicochemical conditions and hydrology within Jamaica Bay may explain the lack of a role for oysters in sediment N cycling among sites. Although measurements of water chemistry monthly and every two weeks were variable at each site, water chemistry was significantly different across the nutrient-loading gradient, as predicted (Fig. 2). In addition, rates of ammonification, nitrification, and DNP varied among sites (Figs. 3 and 4). However, it is possible that the range of conditions available in Jamaica Bay was not broad enough to encompass the trophic gradient over which oysters' role in N dynamics could be detected. Also, we note our box + cage design may have limited water flow across the sediment surface and thus limited

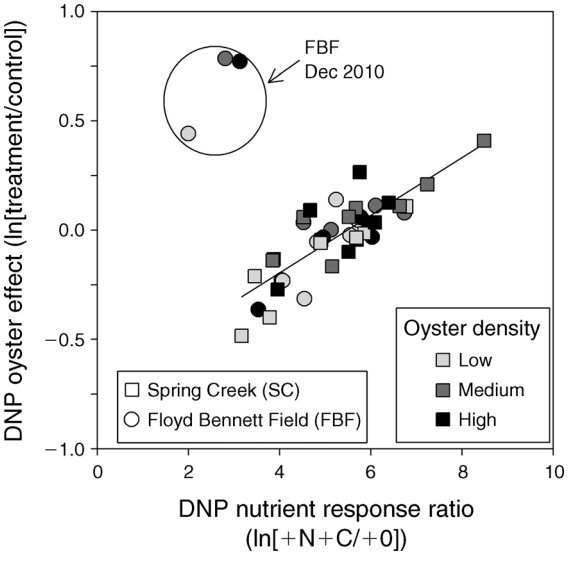


FIG. 5. Simple linear regression between the magnitude of nutrient limitation for denitrification potential (DNP) and the magnitude of the oyster effect on DNP. DNP nutrient-limitation magnitude is expressed as DNP in +nitrate and glucose flasks relative to +no-nutrients flask, while DNP oyster effect is expressed as DNP in oyster treatment sediments relative to DNP in control sediments. Grayscale shading distinguishes oyster density treatments. Results from FBF in December 2010 are outliers (studentized residuals >13.823, $P < 0.001$); with outliers excluded, $r^2 = 0.690$, $P < 0.001$ (with outliers included, $r^2 = 0.013$, $P = 0.464$).

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TABLE 3. Candidate models to estimate nitrification and denitrification potential (DNP) across sites and dates, chosen using Akaike's information criterion (AIC).

Model no.	k†	Likelihood‡	AIC	ΔAIC	R^2	w§	Predictors¶
Nitrification							
1	1	0.00	5642.14	217.15	0.088	0.00	xNH_4^+
2	1	0.00	5621.93	196.93	0.144	0.00	Amf
3	2	0.00	5598.03	173.03	0.211	0.00	Amf, NO_3^-
4	2	0.04	5431.55	6.55	0.533	0.02	Amf, xNH_4^+
5	3	0.02	5432.41	7.41	0.534	0.01	Amf, xNH_4^+, NH_4^+
6	4	0.07	5430.28	5.28	0.540	0.03	Amf, xNH_4^+, NH_4^+, AFDM
7	3	0.15	5428.83	3.84	0.540	0.07	Amf, xNH_4^+, NO_3^-
8	4	0.94	5425.13	0.13	0.548	0.42	Amf, xNH_4^+, NH_4^+, SRP
9	5	1.00	5425.00	0.00	0.551	0.45	Amf(+), xNH_4^+(+), NH_4^+(+), NO_3^-(–), SRP(–)
Denitrification potential, DNP							
1	1	0.00	108.05	71.98	0.044	0.00	AFDM
2	1	0.00	105.01	68.94	0.167	0.00	NO_3^-
3	2	0.00	81.66	45.59	0.245	0.00	AFDM, Chl *a*
4	2	0.00	60.77	24.69	0.310	0.00	NO_3^-, AFDM
5	3	0.00	57.31	21.24	0.323	0.00	NO_3^-, AFDM, NH_4^+
6	3	0.00	52.32	16.25	0.337	0.00	NO_3^-, AFDM, C:N
7	4	0.01	46.20	10.13	0.356	0.00	NO_3^-, AFDM, NH_4^+, C:N
8	4	0.01	45.21	9.13	0.359	0.01	NO_3^-, AFDM, SRP, C:N
9	5	0.06	41.76	5.69	0.371	0.03	NO_3^-, AFDM, Chl *a*, xNH_4^+, C:N
10	5	0.24	38.92	2.84	0.378	0.14	NO_3^-, AFDM, NH_4^+, xNH_4^+, C:N
11	6	0.36	38.11	2.03	0.383	0.22	NO_3^-, AFDM, Amf, NH_4^+, xNH_4^+, C:N
12	6	1.00	36.07	0.00	0.388	0.60	NO_3^-(+), AFDM(+), Chl *a*(+), NH_4^+(–), xNH_4^+(–), C:N (–)

Notes: Models 8–9 (nitrification) and Model 12 (DNP) have the lowest AIC score, indicating the best candidates (ΔAIC < 2). The sign of the relationship for each predictor is given for the final models.

† The number of predictors.

‡ The probability that the candidate model is the best, given available data.

§ Evidence that the model is best among competing models.

¶ Abbreviations: xNH_4^+, exchangeable ammonium; Amf, ammonification; NO_3^-, water column nitrate; NH_4^+, water column ammonium; AFDM, sediment ash-free dry mass; SRP, water column soluble reactive phosphorus; C:N, carbon-to-nitrogen ratio in sediment; and Chl *a*, water column chlorophyll *a*. Other predictors entered but not included in any of the best models include water column nitrite, dissolved inorganic nitrogen, and nitrification (for DNP models only).

the horizontal flux of solutes at all four sites. Under more ideal circumstances our experiment would span a larger range of nutrient loading and more effectively document the influence of local hydrologic variation.

Recent studies have shown that oysters can affect denitrification, although the mechanisms are not always clear. For example, Piehler and Smyth (2011) and Kellogg et al. (2013) found higher net N_2 production at oyster reefs relative to other habitats, but the source of N for denitrification or oysters' effect on sediment C was not measured. Newell et al. (2002) showed that when organic matter consisting of simulated bivalve biodeposits was added to well-oxygenated sediments, coupled nitrification–denitrification increased. However, under anaerobic sediment conditions, the dominant flux of N was NH_4^+ release and denitrification was inhibited from lack of nitrification and low NO_3^- diffusion to sediment microbes (Newell et al. 2002). Finally, Carmichael et al. (2012) derived potential oyster-mediated N removal by combining oyster growth data with the Newell et al. (2005) estimate of 0.75 g N·[g oyster dry mass]$^{-1}$·yr^{-1} removal through denitrification and burial. They found that oysters could enhance N removal through denitrification but acknowledged the need for system-specific measurements. Overall, there is no consensus on whether oysters affect denitrification through direct or coupled pathways, or under what

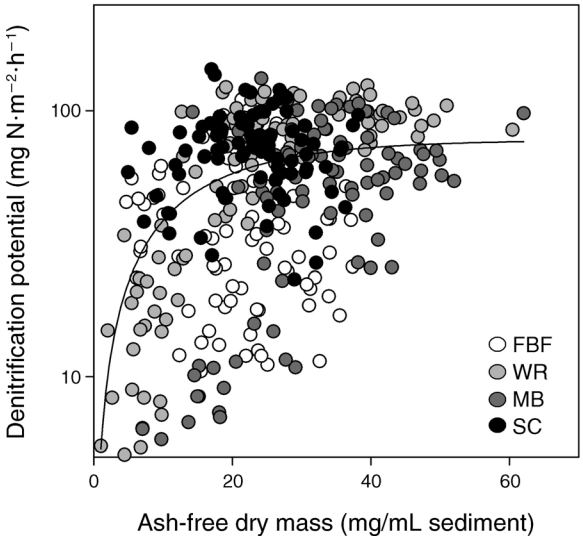


FIG. 6. There was a positive, curvilinear relationship between sediment ash-free dry mass (AFDM) and denitrification potential (DNP). The sites are: FBF, Floyd Bennett Field; WR, Wildlife Refuge; MB, Mott's Basin; and SC, Spring Creek. Note the *x*-axis log scale.

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conditions these effects may change (Newell et al. 2005). These studies show the need to separate direct and coupled denitrification measurements in oyster reefs and adjacent estuarine habitats, and demonstrate that concurrent measurements of environmental drivers of N cycling, especially NO_3^-, carbon, and dissolved oxygen (DO), will be required for future analyses that follow N and C in oyster biodeposits through multiple transformations.

Decoupling of nitrification and denitrification

Strong coupling of nitrification and denitrification is commonplace in marine sediments, but there are conditions whereby the processes become decoupled, as documented at our study sites and other locations (Risgaard-Petersen 2003). Environmental factors that might decouple the rates include low DO at the sediment– water interface (SWI) and high water-column NO_3^-. In addition, sediment porosity and measurement of denitrification with the acetylene-inhibition technique may obscure these relationships. Below we explore each possibility.

Nitrification rates across dates and sites were variable, ranging from 0 to 12 mg $N{\cdot}m^{-2}{\cdot}h^{-1}$, although the median was 0.630 mg $N{\cdot}m^{-2}{\cdot}h^{-1}$ and 80% of rates were <2.0 mg $N{\cdot}m^{-2}{\cdot}h^{-1}$. These results fall within the range reported for the northeast and mid-Atlantic United States, including 0.393 mg $N{\cdot}m^{-2}{\cdot}h^{-1}$ in Chesapeake Bay (Kemp et al. 1990) and 0.958 mg $N{\cdot}m^{-2}{\cdot}h^{-1}$ in Narragansett Bay (Seitzinger and Pilson 1984). This is consistent with data from other biomes, and Herbert (1999) noted an unexpected uniformity in nitrification rates in coastal sediments worldwide. No seasonal patterns in nitrification were apparent from our results. Previous studies of seasonality in nitrification also show no consensus, and have documented both summertime nadirs (Kemp et al. 1990) and peaks (Herbert 1999). We note that rates in the Jamaica Bay were much lower than nitrification measured in eutrophic freshwater sediments that used identical methodology (i.e., nitrapyrin inhibition; Levi et al. 2013).

We did not measure DO at the SWI or in the sediment, but periodic anoxia is common in eutrophic coastal environments and can inhibit nitrification (Cornwell et al. 1999, Jantti and Hietanen 2012). Anoxic conditions in the water column have been reported in Jamaica Bay, especially in summer (Franz and Friedman 2002, Benotti et al. 2007). In addition, the highest nitrification rates occurred at FBF, the site with lowest water-column nutrients and sediment AFDM. This site is close to the ocean inlet, where anoxia at the SWI was less likely. For a concurrent study, we sampled invertebrates from our sediment boxes. The ongoing analysis showed taxa tolerant of low DO were abundant at SC, MB, and WR, with few low DO-tolerant taxa at FBF (C. B. Zarnoch and T. J. Hoellein, *unpublished data*). Overall, we concluded that low DO was likely one limiting factor for nitrification that varied across sites, and could limit transformation of organic and inorganic N from oyster biodeposits into NO_3^-.

A meta-analysis by Seitzinger et al. (2006) showed that coupled nitrification–denitrification was less important as NO_3^- concentrations increase, where the transition began above 10 μmol NO_3^-/L (140 μg N/L). Our study sites showed values >140 μg NO_x-N/L for 81% of the time in SC, 46% and 48% of the time in MB and WR, respectively, and 33% of samples in FBF. In addition, NO_3^- was below detection in exchangeable pore water. Under these conditions, denitrifiers are more likely to use water column NO_3^- than NO_3^- derived from nitrification or in sediment pore water. Seitzinger et al. (2006) also noted that the rate of NO_3^- diffusion into sediments is a controlling factor for direct denitrification. At our study sites we expect that high-NO_3^- seawater was exposed to the sediments as tidal flows moved overlying water across the SWI at regular intervals. However, we note that our box + cage design, in addition to use of sand substrate, could potentially alter fluxes of NO_3^- and DO to the SWI, thereby affecting denitrification rates.

An important consideration for interpretation of DNP results is that the acetylene block technique is a bioassay that measures the ability of the microbial community to conduct denitrification under ideal conditions (Murray and Knowles 1999, Groffman et al. 2006). In the context of this experiment, these data reflect *relative* differences in microbial communities to carry out denitrification after exposure to varying densities of oysters at multiple sites. These data do not reflect in situ rates, and the results should not be used to estimate N_2 flux from sediment. A major benefit of DNP measurements is the capacity for high replication (e.g., we completed 125 separate DNP incubations per day) relative to isotope-tracer methods using flow-through chambers (Cornwell et al. 1999, Groffman et al. 2006). Patterns of DNP rates and limitation in this experiment can be used to isolate treatments (e.g., oyster density 150 oysters/m^2) and sites where oysters are most likely to influence N and C dynamics, which can then be documented with methods to measure in situ N_2 flux (Piehler and Smyth 2011, Kellogg et al. 2013). These analyses have not yet been completed in urbanized ecosystems.

We chose to use a uniform substrate across sites and dates, which has implications for data interpretation and transferability of results. Coarse sand in sediment boxes could increase water column diffusion and promote direct denitrification, however, our data suggest this was minimized. We found relatively high exchangeable NH_4^+ and undetectable pore water NO_3^- in the sediment, which suggested anoxia near the SWI rather than water column diffusion into sediments. In addition, the initial sediment organic-matter content of 0.11% increased to 0.73% in the first two months, and the mean among sites and dates was 1.94%. Finally, organic-matter content in sediment boxes was similar to adjacent sediment. In July

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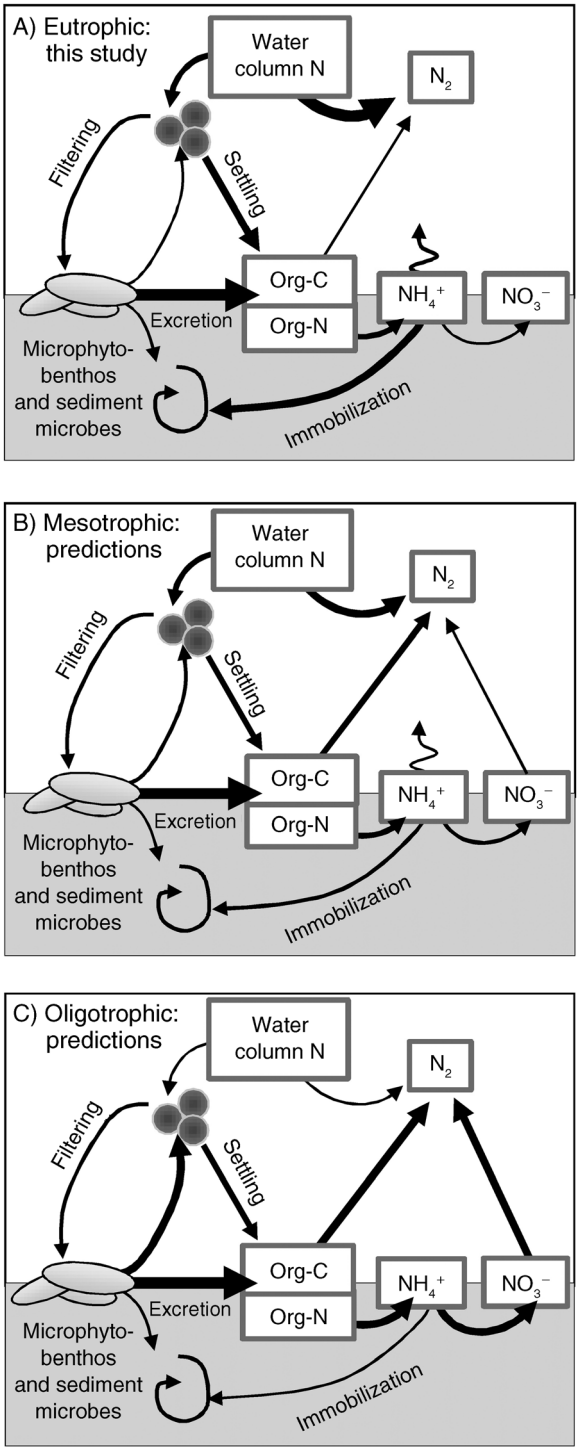


FIG. 7. Conceptual diagram of potential mechanisms whereby oysters influence denitrification in eutrophic vs. oligotrophic locations (modified from Newell et al. [2005: Fig. 2]). Results from our present study suggest the effect in panel A where sediment organic carbon (C) is high, ammonium (NH_4^+) immobilization is preferred over nitrification, and oysters do not affect direct denitrification as it is not primarily C limited. We predict the pathway in panel B could occur in mesotrophic environments where direct denitrification is stimulated by oysters' increase in sediment organic C and water-column nitrate (NO_3^-) is denitrified. The panel C pathway could occur where oysters stimulate coupled nitrification–denitrification and there is minimal interference of nitrification by microphytobenthos or sediment microbes (i.e., nitrification favored over NH_4^+ immobilization).

2010 the percentage organic matter (mean ± SD) in the experimental boxes relative to nearby sediment at Spring Creek was 0.89% ± 0.05% and 0.74% ± 0.05%, respectively, and at Floyd Bennett Field the values were 0.42% ± 0.05% and 0.56% ± 0.06%, respectively. Overall, conditions in boxes could be different from ambient benthic surfaces, and while our data suggest differences were minimized, results should be interpreted in the context of this design.

Oyster-restoration implications

Enhancing denitrification is an ecosystem service commonly cited for support of oyster restoration (Newell et al. 2005). However, studies of oysters' role in N cycling have been conducted using a variety of oyster densities, analytical techniques, field and laboratory experiments, and different measurements of N and C biogeochemistry. DNP rates in this study captured microbial potential for denitrification among treatments and dates, and the associated metrics (i.e., sediment AFDM, nitrification, and pore water NH_4^+) suggested pathways whereby oysters could affect in situ N_2 production in this urban habitat. Previous studies of oysters' role in N dynamics have taken the converse approach, and focused on rates of N_2 production without measuring the other C and N transformations measured here (Newell et al. 2002, Higgins et al. 2013, Kellogg et al. 2013). The high variability in results and methods among previous analyses suggests a need for an organized framework to predict oysters' effects on N and C biogeochemistry across trophic gradients. By assembling our conclusions with literature-based predictions of how oysters might affect sediment C and N dynamics in oligotrophic and mesotrophic waters, we can explore hypothetical sets of conditions that affect the relationship between oysters and N cycling (Fig. 7). We describe three scenarios. First, where oysters' effect on sediment C and N dynamics are restricted to an increase in sediment AFDM (as in this study; Fig. 7A). In the second scenario, oysters increase direct denitrification (Fig. 7B), and in the third, oyster presence increases coupled nitrification–denitrification (Fig. 7C).

In our study, oysters increased sediment organic matter, through biodeposit accumulation or enhanced deposition, with little effect on sediment N dynamics. We concluded this because (1) denitrification was not primarily C limited, and (2) the N in biodeposits most likely did not supply denitrifiers with NO_3^-, as NH_4^+ immobilization was higher than nitrification in most cases. In this scenario, oysters only influenced sediment

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organic C, while the remainder of the elements transformed by oysters into biodeposits cycled among sediment and water column pools. We expect this may occur under eutrophic conditions elsewhere if sediment organic matter is high and nitrification is inhibited. In such locations the effect of oysters on N_2 production may be low. Studies are needed that follow N in oyster biodeposits into nitrification and denitrification products. For now, we conclude that reduced N loads may be a prerequisite for enhanced oyster-mediated N removal in these habitats.

Our data showed the influence of oysters on DNP was greatest when DNP was most strongly nutrient limited (Fig. 5), suggesting that oysters can alleviate N and/or C limitation of DNP under certain conditions. Thus, it is logical to predict a scenario where oyster-mediated increases in sediment organic matter increase direct denitrification. These are locations where organic C limits denitrification and elevated water column NO_3^- serves as a denitrification substrate. Oyster-mediated increases in sediment C could allow water column NO_3^- to enter the N_2 pool, removing it from the ecosystem (Fig. 7B). The N from oyster biodeposits and seston deposition may not be removed from the ecosystem, however, but persist in other water column or sediment N pools. Such a scenario would most likely occur in sandy or rocky sediments where NO_3^- loading is consistent, such as mesotrophic environments or sites with terrestrial or groundwater N inputs.

In the third scenario, oysters can influence coupled nitrification–denitrification to stimulate the transformation of both N and C in oyster biodeposits to N_2 (Fig. 7C). The successful conversion of N from biodeposits into N_2 requires several transformations and environmental prerequisites. Coupling of nitrification and denitrification is typical under conditions with relatively low organic matter and residence times (i.e., oxic sediments), and low competition for NH_4^+ among benthic organisms (Risgaard-Petersen 2003, Seitzinger et al. 2006). This set of characteristics is most prevalent in well-mixed waters where the oxygenated sediments are low in C. Potential locations include reefs in oligotrophic waters or in sediments below or adjacent to off-bottom oyster aquaculture sites with proper tidal flow. In shallow-water sites, where microphytobenthos is prevalent, there may be significant diurnal variation in oyster-mediated coupled nitrification–denitrification due to DO production under illuminated conditions (Newell et al. 2002, Risgaard-Petersen 2003).

In addition to denitrification, other oyster-mediated ecosystem services such as secondary production, N bioassimilation by oysters, and water filtration, are likely to be context dependent. For example, in oligotrophic conditions where N and C in oyster biodeposits could enhance denitrification, oyster water filtration may be a less valued ecosystem service. Under eutrophic conditions, where DNP is less likely to be affected by increased organic matter (Fig. 7A), enhanced filtration and oyster growth (i.e., N assimilation) might be more attainable ecosystem services. For example, N sequestration in oyster tissue and shell reduce water-column N reduction when harvested (e.g., aquaculture; Newell 2004). However, this may be less economically viable where the scale of aquaculture stocks needed to mitigate high N loads is too large to implement (Higgins et al. 2011). Although N sequestration and harvest may not be a solution to mitigate N loads, it may be a useful supplement to other N-management strategies.

Measuring how sediment N-cycling rates vary under different environmental conditions is a major challenge on its own, and overlaying how oysters interact with the N cycle presents additional complexity. For this reason, the effect of oysters on denitrification rates have only recently been assessed (this study; Newell et al. 2002, Piehler and Smyth 2011, Kellogg et al. 2013), and their influence on ammonification and nitrification have only been addressed for the first time here. Smyth et al. (2013) recently reported significant rates of dissimilatory NO_3^- reduction to NH_4^+ (DNRA) at oyster reefs in summer but not in winter in Bogue Sound (North Carolina, USA). Other potentially significant N transformations have yet to be quantified in the context of oyster reefs, such as N_2O emissions and anaerobic NH_4^+ oxidation (annamox).

The objective of our study design was to capture the influence of oysters on sediment N dynamics across a gradient of water-quality conditions at the four study sites. In contrast to our expectations, we did not find a gradient of oyster effect across scenarios A, B, and C described above (Fig. 7). This was probably because all sites were situated within the same eutrophic embayment and the gradient of environmental conditions was not extensive enough. However, variability among sites and dates allowed us to explore how oysters' influence on sediment N and C dynamics changed with environmental conditions. Future studies that quantify the biogeochemical pathways and underlying controls shown in Fig. 7 will enhance the probability that oyster restoration and conservation will provide ecosystem services, including N removal via denitrification, across diverse ecosystem settings.

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